DESIGN MILESTONES

PROJECT ID 46

MILESTONE 1- Research
Understanding Tea Preparation

- Tea Culture
- Brewing Temperature
- Tea Varieties
- Traditional Tea Preparation
- User Habits

Prompt= *"Why do different tea types require different brewing temperatures?"*

AI Role= Clarifier

AI Response Summary=AI explained that different tea varieties such as black tea, green tea, and white tea require different brewing temperatures because of their oxidation levels and processing methods.
Black tea is fully oxidized and typically requires boiling water, while green and white teas are more delicate and should be brewed at lower temperatures.If delicate teas are brewed with boiling water, they can become overly bitter and lose their natural flavor.

Reflection=This information revealed that tea preparation is more complex than simply boiling water.
Different tea types require different brewing conditions, yet most everyday tea appliances treat tea preparation as a single boiling process.
Understanding this difference highlighted the importance of designing a system that respects the traditional preparation methods of different teas.

Design Decision=The product should support multiple tea varieties and their specific brewing requirements instead of simplifying tea preparation into a single boiling function.

- ✓ Designing a system that supports different tea varieties.
- ✗ Designing a generic kettle-like appliance that treats all teas the same way.
- Tea preparation was transformed from a single boiling action into a multi-tea preparation system that respects traditional tea preparation methods

MILESTONE 2- Define
Identifying the Spatial Gap in Tea Consumption

- Tea Tradition
- Tea Consumption Habits
- Mobility
- Temparature Loss
- Thermal Continuity

Prompt=“*Where do people usually drink tea after brewing it?*”

AI Role= Re- Framer

AI Response Summary=AI explained that tea preparation usually takes place in the kitchen because it requires boiling water and brewing tools. However, tea consumption rarely happens in the kitchen. Instead, tea is typically enjoyed in living rooms, bedrooms, or workspaces where people relax, work, or socialize.
During this transition, tea gradually loses its temperature. Additionally, carrying a hot teapot without a stable base can create ergonomic challenges and safety concerns.

Reflection=This insight revealed an important spatial gap in tea consumption habits. Tea is usually prepared in the kitchen but consumed in spaces such as living rooms, bedrooms, or workspaces. During this transition, tea loses its temperature, and carrying the teapot can become uncomfortable or unsafe. This understanding shifted the project focus from simply brewing tea to maintaining thermal continuity across different spaces in the home.

Design Decision=The design should support tea consumption beyond the kitchen by introducing a portable heated base that allows tea to remain warm while being moved between different areas of the home.

✓ Supporting tea consumption across multiple spaces in the home.

✗ Designing a tea appliance that only functions in the kitchen environment.

Tea preparation was transformed from a kitchen-bound activity into a portable tea experience that maintains warmth and comfort across different spaces in the home.

MILESTONE 3- Ideate

Exploring the Tea Preparation System

- Modular System
- Multi-tea Brewing
- Product Mobility
- Portable Heating
- User Interaction

Prompt= *"How can a tea preparation product support different tea types while also allowing the tea to move beyond the kitchen environment?"*

AI Role= Expander

AI Response Summary= AI suggested approaching the product as a multi-component tea preparation system rather than a single appliance. The system could include a kettle for heating water, a transparent glass teapot that allows users to observe the brewing process, and interchangeable infusers designed for different tea varieties.For portability, AI proposed using a thermal plate that stores heat after brewing instead of relying on heavy battery-based heating systems, and suggested integrating recessed push-push handles to allow the heated base to be carried safely while maintaining a minimal form.

Reflection=This exploration showed that the project should not be approached as a conventional tea machine. Instead, the idea evolved into a modular tea preparation system where different components work together.
The concept combines precise tea preparation with portability, allowing tea to be brewed in the kitchen and then enjoyed comfortably in different areas of the home.

Design Decision=

- a kettle for heating water
- a glass teapot that makes the tea visible during preparation
- modular tea infusers designed for different tea varieties
- a portable heated base that stores heat through a thermal plate
- push-push recessed handles that allow the base to be carried safely while maintaining a clean and minimal form

MILESTONE 4- Prototype
Testing the Product System

Prompt= *"Are the recessed handles safe enough for carrying a heated base, and can the thermal plate maintain tea temperature during transportation?"*

AI Role= Critique

AI Response Summary= AI evaluated safety and performance aspects of the portable base. It highlighted that carrying a heated surface requires careful attention to ergonomics and heat transfer.The recessed handles should be positioned away from the heated area and produced from low thermal conductivity materials to prevent heat from reaching the user's hands. AI also suggested using a cast aluminum thermal plate to distribute and retain heat effectively without requiring active electrical heating.

Reflection= This stage helped evaluate the feasibility of the portable base concept. Testing the design from a critical perspective revealed the importance of balancing thermal performance, safety, and usability.The handle system and the thermal plate needed to work together to ensure that the product remains safe, functional, and comfortable to carry.

Design Decision=

- a cast aluminum thermal plate to store and distribute heat
- recessed handles made from heat-resistant material to prevent heat transfer

- User Safety
- Thermal Performance
- Ergonomic Grip
- Interaction Design
- System Testing

✓ Using a thermal plate to store heat / Integrating recessed handles with heat-resistant materials

✗ Exposed metal handles that could transfer heat to the user / Complex portable heating systems requiring batteries

The base evolved from a simple heating component into a safe portable thermal platform that combines heat retention, ergonomic handling, and user interaction.

MILESTONE 5- System Development
Developing the Multi-Tea Brewing System

- Tea Varities
- Brewing Methods
- Infuser System
- Filtration
- Flavor Preservation

Prompt= *"How do preparation methods differ between black tea and delicate teas such as green and white tea?"*
AI Role= Expander
AI Response Summary=AI explained that different tea varieties require different preparation methods. Black tea is typically prepared through continuous contact between tea leaves and hot water, while green and white teas are more delicate and require shorter contact with water to avoid bitterness.

Reflection=This information confirmed that preparing different tea varieties requires more than simply adjusting temperature. The preparation logic itself also differs.
Based on this understanding, the design was developed to include two different infuser systems, each supporting the preparation characteristics of specific tea types.

Design Decision=
- Black tea infuser with stainless steel mesh filter
- Green / white tea infuser with a glass gravity-flow system

Each module supports the traditional preparation logic of its tea type.

✓ Designing two separate infuser systems for different tea varieties
Using stainless steel mesh for black tea filtration
Using glass infuser for delicate teas
✗ Using a single universal infuser for all tea types / Forcing delicate teas into traditional black tea brewing methods

Tea preparation evolved from a single brewing method into a modular multi-tea preparation system.

MILESTONE 6- Product Architecture Structuring the Tea Brewing System

- Product Architecture
- System Integration
- Portable Base
- Tea Preparation Flow
- Modular Components

Prompt= *"How can the kettle, teapot, infusers, and portable base work together as one integrated system?"*

AI Role= Re-Framer

AI Response Summary=AI suggested structuring the product as a multi-component system rather than a single appliance.The system would consist of a heating kettle, a glass teapot, interchangeable tea infusers, and a portable heated base.Each component would have a specific role within the tea preparation process while maintaining visual and functional harmony.

Reflection=This stage helped clarify that the product should not be perceived as a traditional tea machine.
Instead, the design functions as a system of interconnected components, each responsible for a specific part of the tea preparation and serving process.
By separating these roles, the product becomes more flexible and better aligned with real tea consumption habits.

Design Decision=

- Heating kettle for boiling water
- Glass teapot where tea preparation becomes visible
- Modular tea infusers for different tea types
- Portable heated base that maintains temperature outside the kitchen

✓ Designing the product as a multi-component system
Separating brewing, infusion, and serving functions

✗ Designing a single compact tea machine where all functions are merged into one body

The product evolved from a single appliance concept into a modular tea preparation

MILESTONE 7- Material Desicions
Selecting Materials for Performance and Experience

Prompt= *"Which materials can support heat resistance, visual clarity, and durability in a tea brewing system?"*

AI Role= Expander

AI Response Summary=AI explained that different materials can support both functional performance and visual clarity in tea preparation products. Glass allows users to observe the tea during preparation, while porcelain provides heat resistance and a refined surface suitable for kitchen appliances. For filtration, stainless steel mesh offers durability and effective separation of tea leaves. Materials with strong thermal conductivity, such as aluminum, can also support heat distribution and retention within the base.

- **Material Contrast**
- **Heat Resistance**
- **Durability**
- **Visual Clarity**
- **Thermal Performance**

Reflection=This stage highlighted that material choices should support both functional and experiential aspects of the product.The combination of glass, porcelain, and metal elements creates a balance between visual transparency, durability, and thermal performance.These materials also support the cultural perception of tea preparation as a refined and attentive process.

Design Decision=

- Glass for the teapot to allow visual observation of the tea
- Porcelain body for the kettle to provide heat resistance and a refined appearance
- Stainless steel mesh for the black tea infuser to ensure durability and easy cleaning
- Cast aluminum thermal plate within the base to retain and distribute heat
- Heat-resistant handles to allow safe transportation

✓ Glass for visual transparency / Porcelain for heat resistance and refined appearance
Stainless steel mesh for filtration durability

✗ Fully metallic bodies that hide the tea preparation process
Plastic components in heat-sensitive areas

Material selection evolved from purely functional choices into a balanced combination of performance, durability, and visual experience.

MILESTONE 8- Interaction Design
Designing the User Interaction System

- User Interface
- Tea Selection
- Temperature Control
- Information Display
- Ambient Lighting

Prompt= *"How can a user interface help simplify tea preparation while also informing the user about the selected tea?"*

AI Role= Clarifier

AI Response Summary=AI explained that many kitchen appliances simplify complex processes through predefined modes instead of manual technical adjustments. This confirmed that selecting the tea type directly can make the interaction clearer and easier for users.

Reflection= The designer proposed a tea-type based digital interface to simplify tea preparation. AI feedback supported this idea by confirming that predefined modes improve usability and clarity.

Design Decision= The product includes a digital control panel designed to simplify tea preparation.The interface allows users to select:

- Green tea
- White tea
- Black tea

Each selection automatically activates the appropriate preparation conditions.The screen can also display small informational pages about the selected tea, such as preparation temperature and basic characteristics.In addition, the base integrates soft ambient lighting that creates a warm atmosphere during tea consumption.

✓ Tea-type based interaction instead of manual temperature selection / Informational interface that communicates tea characteristics / Ambient lighting integrated into the base

✗ Complex technical control panels with multiple confusing settings

✗ Interfaces that only display temperature values without context

The interface evolved from a simple control panel into a guided tea preparation interface that supports both usability and user awareness.

MILESTONE 9- Final Design Tranformation From Initial Idea to Final Product

Prompt= *“How can all components of the tea preparation system work together to support both brewing accuracy and tea culture?”*

AI Role= Critique

AI Response Summary=AI reviewed the overall system and highlighted the importance of balancing technical performance, usability, and cultural context within a single product. This confirmed the need for a design where all components work together as a coherent system.

Reflection= The designer developed the product as a complete tea preparation system rather than a single appliance. AI feedback helped confirm that integrating brewing accuracy, portability, and user interaction strengthens the overall product concept.

Design Decision=

The final product integrates:

- A porcelain kettle for boiling water
- A transparent glass teapot that makes the tea visible
- Two specialized infusers for different tea types
- A portable heated base that maintains temperature outside the kitchen
- A digital interface that guides the user during preparation

- **Design Evolution**
- **System Integration**
- **Tea Culture**
- **Portable Experience**
- **Product Identity**

✓ Designing the product as a complete tea preparation system

✗ Treating tea preparation as a simple boiling task

The project evolved from a basic tea appliance concept into a portable tea preparation system that supports tea culture and modern usability.